- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

13 The displacement $\mathbf{r}$ m of a parachutist t s after opening their parachute is modelled by

$$\mathbf{r} = \begin{pmatrix} 50t \\ 280 + 5t - 280e^{-0.16t} \end{pmatrix}$$

where the x-direction is horizontal and the y-direction is vertically **downwards**.

(a) Calculate the **distance** from the parachutist's initial position that the model predicts after 10 s. **[3]**

(b) Find a vector expression for the velocity of the parachutist according to the model. **[3]**

(c) Determine what velocity the model predicts for large values of t. **[2]**

(d) Parachutists usually land travelling approximately vertically.

Explain a factor that should be included in the model to better reflect this. **[1]**

Turn over for question 14